Nathan de Castro

Full-Stack Developer & Solution Architect

Languages — French · English (Native)

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PROFILE

Full-stack developer and solution architect with **8 years of experience** shipping production software end to end: web platforms, mobile apps, real-time backends and interactive installations running on Android, iOS, Windows and Linux. Owned systems from the first workshop with domain experts through to the deployed, maintained product: architecture, implementation, infrastructure, hardware selection, deployment and support. Heavy daily user and integrator of LLM and agentic AI tooling across the development lifecycle. Comfortable running the technical conversation and the client conversation in the same room.

WORK EXPERIENCE

Co-Founder & Lead Engineer — Valbilon Studios

Sion, Switzerland · 2022 – Present

Principal developer across the company's entire product portfolio: Flutter mobile apps, web platforms, Unity installations and Windows/Linux kiosk software, owning architecture, implementation, deployment and support.

- **CPPS digital suite**: designed and built the full technology stack for a public earthquake training centre and museum, a reservation and payment platform for state, private and educational clients, with automated invoicing, accounting and reporting. **50,000 users/year**. (REST APIs, Stripe, Firebase)
- **CPPS exhibit and kiosks**: built the centre's integrated exhibit framework, home to **Europe's largest earthquake simulator** across 3 spaces and 27 interactive stations, designed with seismology experts and university professors. **30+ touchscreen mini-games** and a fleet of **30 wall-mounted tablets**, each running its own kiosked webapp/APK over a secured local network with Docker, centrally configured and remotely updated. **30,000 visitors/year**. (Android, Docker, kiosk and touchscreen interfaces)
- **Escape With Wallis**: hour-long Android escape game guiding visitors through the exhibit, with offline-tolerant progress sync and trigger-driven scripted events. **3000+ sessions completed**. (Android, Firebase)
- **Winston**: architected and built the company's finance and accounting platform. LLM-driven document ingestion, categorisation and reconciliation on a conventional ledger core, with invoicing / VAT / payroll / Business expenses / AI automated cold calling. Replaced **4+ separate SaaS subscriptions** and still expanding. (LLM API integration, agentic workflows, .NET)
- **Exploreum**: multi-tenant platform connecting museums with cross-institution progress tracking and rewards; currently in rollout. (Firebase, REST APIs)
- Designed and rolled out agentic AI workflows across multiple delivery pipelines (specification, scaffolding, code review, testing and documentation) (Claude Code, MCP)

Technical Lead — Surgical VR (formerly MedicalVR)

Amsterdam, Netherlands · 2022

- **PulmoVR**: developed and maintained a VR clinical application to visualise complex anatomical structures and accelerate lung segmentectomy planning through machine-learning segmentation. Clinically validated results: **-30 min** average operating time for 3D VATS vs. 2D VATS, **>97%** anatomically accurate 3D models, and a **52%** surgical plan change rate after 3D model review. (C#, Unity)
- Led delivery for the development team within an Agile/Scrum framework: sprint planning, daily stand-ups, task breakdown, delegation, code reviews and change documentation, tracked via GitHub Projects.
- Drove **CE marking** and **ISO 27001** readiness: information-security risk assessment and mitigation, controlled change management, and the technical documentation required for a regulated medical product.
- Ran requirements gathering and validation directly with surgeons and clinical staff, then fed their feedback back into the implementation.

Senior Software Engineer — The Barn

Delft, Netherlands · 2019 – 2021

- **Eventrum**: built a browser-based virtual conference platform on WebRTC, delivered in partnership with **TU Delft and Accenture**, with real-time audio/video, room signalling and presence for up to **1k concurrent participants**. (WebRTC, Unity, JavaScript)
- **TeamUp**: networked multiplayer team-building game delivered to WebGL and Windows, with authoritative session state over Photon. (WebGL, Photon, Unity)
- **BMO industrial simulator**: built a virtual training simulator for the client's industrial sorting machine, with a scenario-authoring tool trainers used to create build, maintenance and operations exercises, cutting the cost and time of flying in a senior trainer for **2-week onboarding trips**. Deployed on WebGL and Windows at the client's facility. (C#, Unity)
- Led within an Agile/Scrum team: sprint planning, daily stand-ups and code reviews, tracked via ClickUp.

SELECTED FREELANCE WORK & OPEN SOURCE

SHIPPED APPS

- **Spinlab** (Flutter, Firebase, Stripe): mobile app for iOS & Android connecting tennis coaches and students, with location-based booking and in-app payments handled end-to-end.
- **Sandlog** (Flutter): dream logging and tracking app.
- **Pygmak** (Unity): mobile game with micro-transactions.

PROPRIETARY / CLIENT WORK

- **Badger** (Flutter): virtual business card app built for conventions and networking events.
- **ColisMarket** (Flutter): fully functioning prototype built for the Tunisian parcel-delivery market, handling parcel delivery path tracking with automated OCR parcel-ticket reading and filing, ready but awaiting market launch.

Other open-source packages, tools and plugins are available on GitHub.

EDUCATION

Research in Game Development
University of Technology Sydney, 2018 accelerated program
First Class Honours
Computer graphics implementation, VR applications for the handicapped, and teaching first-year students in game development

BSc Computer Science
Goldsmiths, University of London, 2017
First Class Honours
Machine learning, artificial intelligence, advanced graphics, audio-visual processing

BA Media, Music & Sound
Edge Hill University, 2014
First Class Honours

Baccalauréat Scientifique
(Physics & Mathematics), Lycée Français Vincent Van Gogh, 2013

CERTIFICATES

C++ Programming — EPFL **C# Masterclass** — Udemy

TECHNICAL SKILLS

Languages	C#, .NET, Dart, JavaScript, TypeScript, Python, Lua, SQL, HTML, CSS
Frontend & Mobile	Flutter, Android, iOS, WebGL, Unity, responsive UI, kiosk and touchscreen interfaces
Backend & Cloud	Firebase (Authentication, Firestore, Cloud Functions, Storage), AWS, REST APIs, Stripe, in-app purchases, WebRTC, Photon, real-time synchronisation
Platforms & Ops	Windows, Linux, Android device fleet management, Git, CI/CD, Docker, app store release management, remote deployment and monitoring, Agile, Scrum, Kanban
AI	LLM API integration, retrieval-augmented generation (RAG), agentic workflows, prompt engineering, Claude Code, GitHub Copilot, MCP